



Research paper

Evaluation of groundwater suitability for domestic and agricultural utility in semi-arid region of Anantapur, Andhra Pradesh State, South India



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ABSTRACT

This study reports the evaluation of groundwater suitability for domestic and agricultural utility in semi-arid region of Anantapur, Andhra Pradesh State, South India. Fifty groundwater samples were analysed for parameters such as pH, electrical conductivity, total dissolved solids, total alkalinity, total hardness, bicarbonates, calcium, magnesium, nitrate, sulphate, phosphate, sodium, potassium and fluoride. The analysis results shows that the concentration of fluoride in 42 samples out of 50 exceed the limit for fluoride (1.5 mg/L) in drinking water set by the W.H.O. Based on the WQI shows overall quality of groundwater is not suitable for drinking purpose. It is inferred from the study that these water sources can be used for potable purpose only after prior treatment. The parameters such as sodium adsorption ratio, per cent sodium, residual sodium carbonate, were calculated for the evaluation of irrigation water quality. Groundwater chemistry was also analysed by statistical analysis, USSS, Wilcox and Piper diagrams, to find out their suitability for irrigation and the analysis reveals that most of groundwater samples of the study area fall under excellent class suitable for irrigation.

1. Introduction

Groundwater is well known as a source of potable water since time immemorial. The other merits of groundwater over surface water are little need of treatment extensive storage potential of aquifers, annual replenishment by rainfall, and decentralized availability close to demand centers saving transport costs. The groundwater quality problems are due to contamination and over exploitation. The quality of groundwater gradually changes with respect to depth, seasonal changes, and due to the types of rock present in an area.

Quality of groundwater is as important as the quantity. Groundwater quality data provides us the geological history of the rocks, groundwater recharge, discharge, movement and storage (Walton, 1970). Now a days the demand for fresh water is increasing due to increase of population, urbanization, industrialization and intense agricultural activities in many parts of the world (Raju et al., 2011). Unjustified, unreasonable groundwater development and management in the country owing to urbanization, industrialization and population demands for food security is changing hydrological and geochemical environment of aquifers causing groundwater pollution.

Groundwater forms a major source for drinking water in urban as well as in rural areas. More than 90% of the rural population uses groundwater for domestic purposes (Sunitha et al., 2012). The quality

of ground water is of great importance in determining the suitability of ground water for a certain use (public water supply, irrigation, industrial applications etc). The water used for drinking purpose should be free from any toxic elements, living and nonliving organisms and excessive amount of minerals that may be hazardous to health (B.M Reddy et al., 2013).

The quality of drinking water has increasingly been questioned from health point of view for many decades (Vinod et al., 2009). The alarming rate of population growth, evolving industrial society, advances in technology, and the existing trend of depletion of groundwater resource has raised some serious environmental problems. Groundwater quality studies with reference to drinking and irrigation purposes in different regions have been carried out, viz., Anantapur (Sunitha et al., 2012, 2014), Kadapa (Sunitha et al., 2013, 2016) Tamil Nadu (Subramani et al., 2005), Uttarakhand (Jain et al., 2010), Punjab (Kumar et al., 2007), Sri Lanka (Nishanthiny et al., 2010), Iran (Aghazadeh and Mogaddam, 2010; Narany et al., 2015), Nigeria (Ackah et al., 2011), and China (Zhang et al., 2012). Groundwater in India accounts for about 80% of domestic water requirements and more than 45% of total agricultural water, irrigating 39 million hectares (Kumar et al., 2005). Groundwater is also the single largest and most productive source of water for irrigation in India (Kinzelbach et al., 2003). Mikayilov and Acar (1998) has concluded that the relationship between

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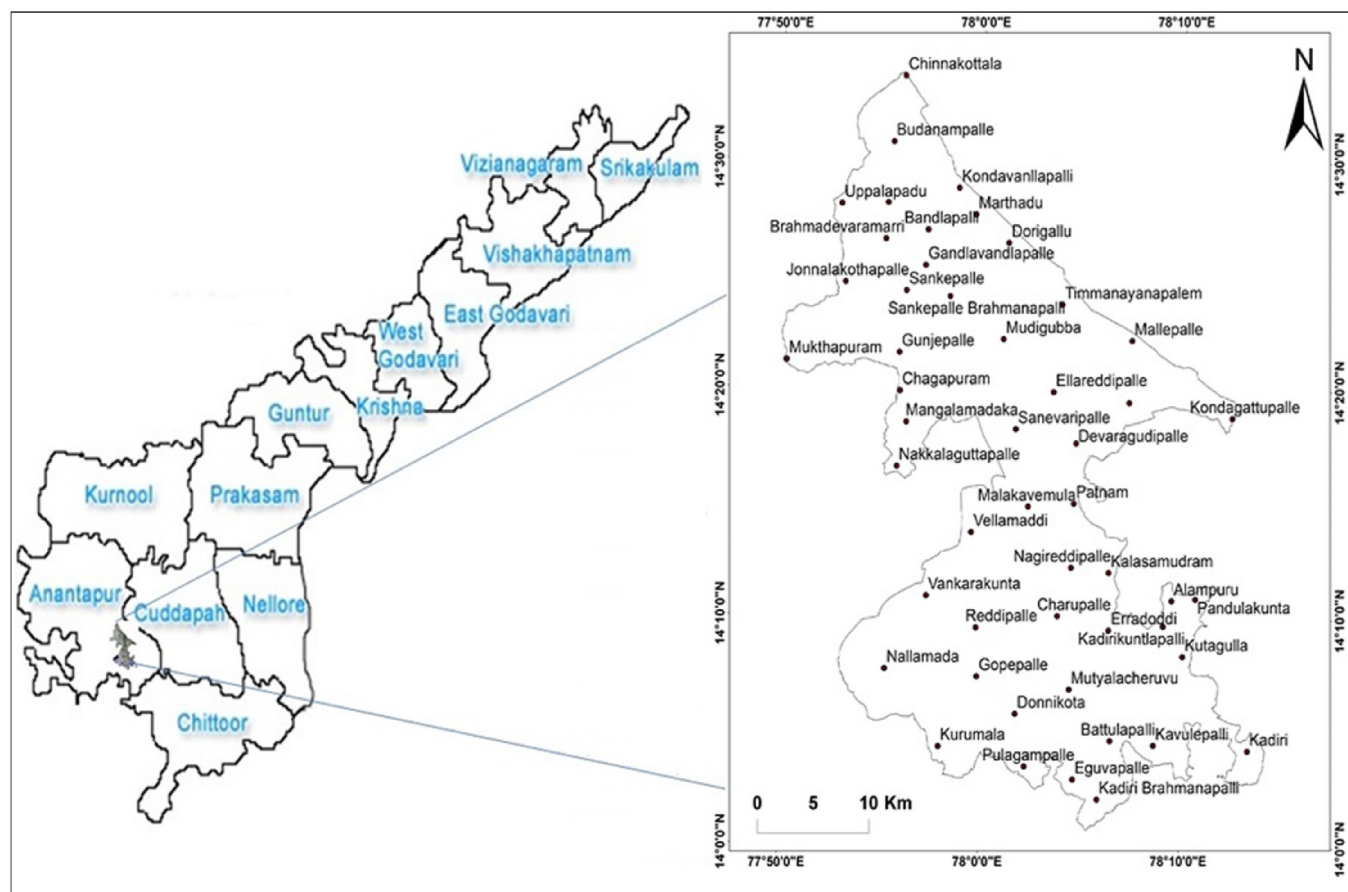


Fig. 1. Location map of the study area.

agricultural practices and groundwater pollution was established. Earlier studies have identified different types of groundwater pollution like geogenic or anthropogenic source, major ions, nitrogen pollutants, trace elements, agricultural pesticides, dissolution or weathering of the rocks and soil (Adimalla and Venkatayogi, 2017; Narsimha et al., 2017a, b; Narsimha et al., 2013a, b; Ding and Zhang, 2002; Islam and Singhal, 2004; Elhatip et al., 2003; Kara et al., 2004; Krishna and Govil, 2004; Narsimha et al., 2013a, 2013b; McBride, 2004; Guo et al., 2007).

2. Study area

The present study area is located in the southwestern part of the Anantapur district and forms part of the Survey of India toposheet Nos: 57 F/14, F/15, F/16, J/3, J/4 and lies between the North latitudes $77^{\circ}15'0''$ – $78^{\circ}50'0''$ and East longitudes $14^{\circ}0'0''$ – $14^{\circ}35'0''$ covering an area of 1136.3 km^2 . Location map of the study area is shown in Fig. 1. The major litho units observed are peninsular gneisses of Archean age consisting of pink granites, schists, composite gneisses of Dharwar age, intruded by a few pegmatite dykes. The major geomorphic units of the study area are mainly denudational Hills, dissected pediments, pediplain and alluvium (Sunitha et al., 2012). The study area is drained by Chitravathi River and its tributaries. Chitravathi River is a fifth order of streams with dendritic drainage pattern and flows from South to North. The area experiences a semiarid climate with a moisture index of 33.7% with mean monthly temperatures of 17°C in January to 42°C in May. The soils of the study area are predominantly of the black and red types. The soils of the area are derived from granitic and its associated rocks and are generally alkaline (with a soil pH of about 9). Total annual rainfall is 560 mm.

Groundwater in Achaean crystalline rocks like granites, gneisses and

Dharwarian schists occurs in weathered and fractured zones. These rocks have developed secondary porosity. The degree of weathering in these formations is less than 20 m and this zone is tapped extensively by the dug well and dug cum bore wells. The depth of open wells varies from 6.0 to 25.0 m below ground level (bgl) and depth to water level varies from 1.5 to 23 m bgl. The yield of dug wells varies from 10 to 200 L/day for a pumping period of 3–6 h a day (CGWB, 2012).

3. Materials and methods

Fifty water samples are collected in pre monsoon during April 2017. The samples are collected in pre-cleaned polyethylene bottles of 2 L capacity after rinsing to stabilize the electrical conductivity (Khairwal and Garg, 2006). The groundwater samples are analysed as described by American Public Health Association (APHA, 1995) procedure, and suggested precautions are taken to avoid contamination. The various parameters determined are pH, electrical conductivity, total dissolved solids, total hardness, calcium, magnesium, total alkalinity, carbonate, bicarbonate, chloride, sulphate, sodium, potassium, nitrate and fluoride. pH and EC are determined by pH meter, conductivity meter, TDS are determined by indirect method (Raghunath, 2003), Total Hardness, Ca^{2+} , Mg^{2+} , CO_3^{2-} , HCO_3^- and Cl^- are determined by titrimetry, whereas Na^+ and K^+ are determined by flame photometry (Systronic Model No.128), SO_4^{2-} and NO_3^- are determined by spectrophotometric method. F^- is determined by using ion selective electrode (Orion 4 star ion meter, Model: pH/ISE). The depth of the bore wells varied between 250 and 700 feet. Water flows in the direction of the steepest downhill gradient. Flow direction was calculated for each pixel using the filled DEM, i.e. the direction in which water will flow out of the pixel to one of the eight surrounding pixels. Arc Hydro in Arc

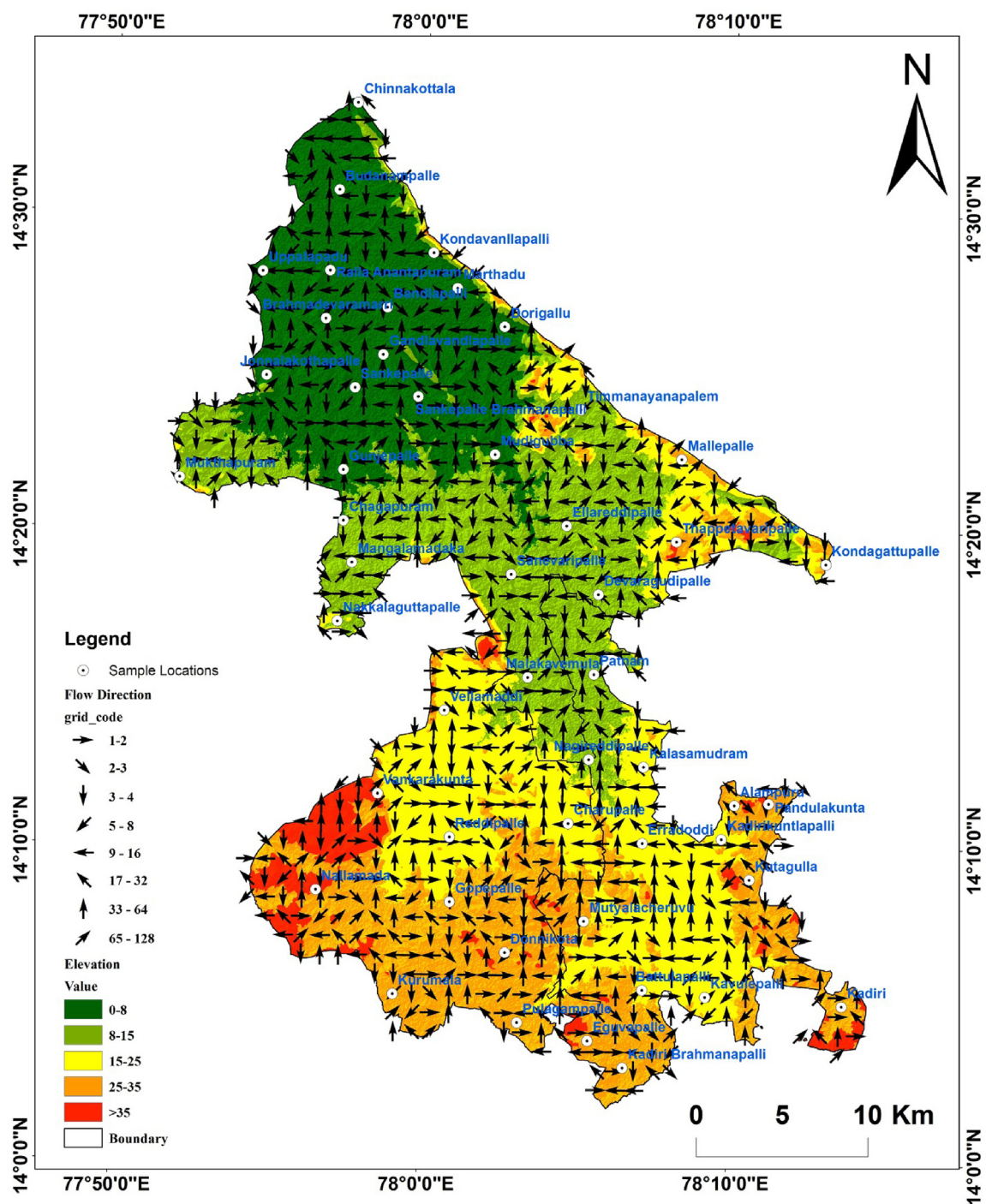


Fig. 2. Groundwater flow direction map of study area.

Table 1

Instrumental, titrimetric and calculation methods were used for chemical analysis of groundwater samples from Anantapur, Andhra Pradesh State.

Parameter	Method	Instruments	Unit	References
pH	Digital pH Meter	pH Meter (Systronics MK VI)	–	APHA (1999)
EC	Digital Conductivity Meter	Conductivity Meter (Systronics Model No-304)	μs/cm	Hem (1991)
TDS	Indirect method (Raghunath, 2003)	$0.64 \times EC \mu S/m$	mg/L	APHA (1999)
Sodium & Potassium	Flame Photometry	Flame Photometer (Systronics Model No-128)	mg/L	APHA (1999)
Calcium & Magnesium	Volumetric	Titration	mg/L	APHA (1999)
Chloride	Volumetric	Titration	mg/L	APHA (1999)
Bicarbonate	Volumetric	Titration	mg/L	APHA (1999)
Sulphate & Nitrate	Turbidity	Spectrophotometer	mg/L	APHA (1999)
Fluoride	Ion selective electrode	Orion 4 star ion meter, Model: pH/ISE	mg/L	APHA (1999)

Table 2
Analytical data for the groundwater samples from the study area.

Sample ID	Longitude	Latitude	pH	EC $\mu\text{S}/\text{cm}$	TDS mg/L	TH mg/L	Na ⁺ mg/L	K ⁺ mg/L	Ca ²⁺ mg/L	Mg ²⁺ mg/L	HCO ₃ ⁻ mg/L	SO ₄ ²⁻ mg/L	Cl ⁻ mg/L	F ⁻ mg/L	NO ₃ ⁻ mg/L	%Na	RSC	SAR
MGW-1	77°94'17.23"E	14°45'47.79"N	7.5	470	300	580	150	28	55	32	95	150	130	4.8	60	57.3	-3.8	2.3
MGW-2	78°07'94.57"E	14°39'61.40"N	7.3	460	294	590	120	27	60	48	100	160	120	4	50	45.9	-5.3	1.8
MGW-3	77°99'09.67"E	14°40'59.74"N	7.5	440	281	600	150	26	55	68	120	170	140	3	52	46.3	-6.3	2.2
MGW-4	78°05'35.80"E	14°26'99.00"N	7.5	430	275	620	130	26	58	92	90	140	130	3.2	55	37.6	-8.9	2.8
MGW-5	77°96'70.51"E	14°49'33.03"N	7.7	510	326	610	190	14	56	65	90	130	110	4.9	56	51.4	-6.6	2.9
MGW-6	78°12'48.40"E	14°34'12.74"N	7.9	520	333	380	150	12	60	75	80	120	90	2	80	42.7	-7.8	2.7
MGW-7	77°61'70.45"E	14°43'50.12"N	8.1	530	339	370	120	12	40	50	80	110	100	4.7	82	47.4	-4.7	2.7
MGW-8	77°96'24.89"E	14°47'11.05"N	8.2	490	314	370	130	25	38	50	85	120	120	4.6	86	51.1	-4.6	2.2
MGW-9	78°07'14.51"E	14°23'59.95"N	8.4	500	320	360	200	16	35	48	85	90	130	3.8	82	61.5	-4.3	4
MGW-10	77°96'22.32"E	14°43'14.07"N	8.6	480	307	400	140	14	40	50	80	90	140	0.9	72	51.3	-4.7	2.2
MGW-11	78°08'31.83"E	14°33'04.59"N	8.6	460	294	410	130	12	50	45	75	120	160	2.6	75	49	-4.9	2
MGW-12	77°97'51.49"E	14°45'46.90"N	7.4	440	281	420	140	14	50	50	70	130	170	5.7	60	49.3	-5.4	2
MGW-13	77°99'84.95"E	14°44'68.36"N	7.6	450	285	430	100	14	50	75	65	110	150	2.4	65	35.2	-7.5	1.5
MGW-14	78°00'45.72"E	14°38'91.91"N	7.8	410	265	440	100	22	45	75	70	90	140	4.7	50	36.8	-7.2	1.5
MGW-15	78°00'23.88"E	14°35'21.72"N	7.9	430	275	470	90	15	45	65	72	90	170	2	57	36.1	-6.4	1.4
MGW-16	78°14'87.60"E	14°33'73.56"N	8	530	339	500	90	14	46	60	74	110	180	1	46	37.1	-6	1.5
MGW-17	77°96'03.19"E	14°35'17.31"N	8.2	510	326	510	80	12	36	60	76	120	200	1.2	52	36	-5.4	1.5
MGW-18	77°96'98.66"E	14°33'46.50"N	8.4	520	333	520	140	11	38	83	78	100	220	2.4	52	42.2	-7.4	2.6
MGW-19	77°91'24.44"E	14°35'04.23"N	8.6	550	352	530	150	16	42	92	82	90	160	2	54	41.7	-8.3	2.5
MGW-20	77°97'82.55"E	14°31'09.08"N	8	540	346	540	70	16	44	65	88	95	170	4.6	56	31.4	-6.1	1.1
MGW-21	78°08'30.09"E	14°30'43.59"N	8.2	550	352	580	90	14	48	60	92	85	180	1.3	65	36.8	-5.8	1.5
MGW-22	77°92'81.93"E	14°34'95.13"N	7.2	570	365	600	70	11	50	40	96	80	150	4.6	70	36.5	-4.2	1.1
MGW-23	77°91'87.77"E	14°35'39.76"N	7.4	590	378	610	100	10	55	35	110	70	160	3.9	74	45	-3.8	1.5
MGW-24	78°11'09.87"E	14°36'99.55"N	7.6	610	390	620	90	14	55	35	120	75	170	2	74	43.1	-3.6	1.4
MGW-25	77°95'56.62"E	14°52'75.01"N	7.8	510	326	630	100	13	56	50	100	100	150	3	75	40.4	-5.2	1.9
MGW-26	77°96'31.51"E	14°34'95.71"N	8	530	339	640	100	15	60	46	90	110	140	2	77	41.1	-5.3	1.7
MGW-27	77°96'46.59"E	14°27'91.01"N	8.2	550	352	650	75	10	65	48	90	120	180	2.2	78	32.8	-5.7	1.2
NGW-28	78°07'40.44"E	14°07'84.19"N	8.4	570	365	660	80	10	65	75	100	130	190	3.6	82	28.4	-7.7	1.2
NGW-29	78°00'52.20"E	14°10'91.31"N	8.6	590	378	680	100	11	66	80	120	140	200	1.2	88	31.9	-7.9	1.6
NGW-30	78°03'61.06"E	14°17'67.08"N	8.8	610	390	730	120	12	68	85	130	150	210	4.5	75	34.7	-8.2	1.7
NGW-31	77°98'18.60"E	14°14'17.63"N	7.8	620	397	580	95	25	70	85	110	160	220	2.2	70	31.2	-8.6	1.3
NGW-32	78°01'29.56"E	14°09'54.02"N	8	640	422	600	70	28	80	80	100	140	230	3	65	26.2	-8.9	1.2
NGW-33	78°05'87.48"E	14°10'90.15"N	8.2	660	410	620	75	36	70	75	100	130	240	5.2	70	30.2	-8	1.2
NGW-34	78°03'60.28"E	14°23'38.34"N	8.4	670	435	480	70	14	50	80	95	120	270	3.5	65	27.2	-7.5	1.2
NGW-35	78°03'65.19"E	14°13'10.58"N	7.2	650	416	440	80	12	52	80	90	110	280	3.4	70	29.2	-7.7	1.5
NGW-36	78°09'45.15"E	14°16'80.19"N	7.4	490	314	430	100	15	54	74	95	90	300	1.2	60	35	-7.2	1.1
NGW-37	77°99'14.74"E	14°18'61.12"N	7.6	510	326	450	75	11	56	83	120	80	270	2.3	55	26.9	-7.6	1.3
NGW-38	78°04'73.51"E	14°31'42.84"N	7.8	530	339	500	90	10	60	57	130	85	250	2.2	50	35.1	-5.5	1.5
NGW-39	77°99'02.37"E	14°11'27.60"N	8	550	352	470	130	13	62	62	120	180	260	2	45	42.2	-6.2	2.1
KGW-40	78°16'03.87"E	14°11'31.30"N	7.6	560	358	460	140	13	50	66	100	190	270	2.2	50	44.7	-6.2	2.2
KGW-41	78°09'31.18"E	14°23'38.40"N	7.8	500	320	470	120	15	48	65	90	200	280	2.5	40	41.9	-6.2	1.7
KGW-42	78°13'93.17"E	14°06'72.88"N	8	480	307	500	150	16	50	66	100	220	300	2.4	40	46.6	-6.2	2
KGW-43	78°17'93.36"E	14°16'95.46"N	7.2	460	294	520	130	12	52	66	90	240	280	4.3	40	42.6	-6.5	1.7
KGW-44	78°18'62.25"E	14°18'71.39"N	7.4	480	307	530	150	15	54	57	100	140	300	4.3	35	48.3	-5.7	1.9
KGW-45	78°10'35.99"E	14°19'12.98"N	7.6	510	326	510	130	12	56	57	110	160	280	1.5	32	44.3	-5.6	1.6
KGW-46	78°16'35.39"E	14°17'32.73"N	7.8	530	339	500	100	12	58	64	120	170	300	4.7	45	36.3	-6.1	1.2
KGW-47	78°08'49.38"E	14°10'28.14"N	8	540	346	480	110	12	62	72	100	180	270	1.2	50	36	-7.3	1.3
KGW-48	78°11'03.74"E	4°20'51.71"N	8.2	550	352	470	150	11	60	65	110	190	260	2.5	60	44.9	-6.5	1.6
KGW-49	78°10'46.17"E	14°07'73.81"N	8.4	570	364	460	120	10	56	64	120	200	250	2	65	40.4	-6	1.3
KGW-50	78°17'24.91"E	14°07'27.42"N	8.4	590	378	420	150	12	55	64	130	220	240	2.2	68	46	-5.8	1.5

Table 3
Statistical summary of physico-chemical parameters in the study area (n = 50).

Water Quality Parameters	W.H.O accept limit	BIS accept limit	Minimum	Maximum	Mean	Standard Deviation	Number of samples exceeding the permissible limit	% of samples exceeding the Permissible limit
pH	7.0–8.5	6.5–8.5	7.2	8.8	7.9	0.4226	5	10
EC ($\mu\text{S}/\text{cm}$)	400–1500	500–1500	410	670	528.8	62.682	Nil	Nil
TDS (mg/L)	500–1500	500–1500	265	435	338.4	40.491	Nil	Nil
TH (mg/L)	100–500	300–600	360	730	518.8	91.152	14	28
Na^+ (mg/L)	50–200	–	70	200	114.6	31.474	Nil	Nil
K^+ (mg/L)	200	–	10	36	15.4	5.948	Nil	Nil
Ca^{2+} (mg/L)	75–200	75–200	35	80	53.7	9.456	Nil	Nil
Mg^{2+} (mg/L)	50–150	30–100	32	92	63.68	14.919	Nil	Nil
HCO_3^- (mg/L)	30–150	–	65	130	96.6	17.193	Nil	Nil
SO_4^{2-} (mg/L)	200–400	150–400	70	240	132	42.402	Nil	Nil
Cl^- (mg/L)	200–600	250–1000	90	300	200.2	62.547	Nil	Nil
F^- (mg/L)	0.6–1.5	0.6–1.5	0.9	5.7	2.96	1.314	42	84
NO_3^- (mg/L)	45	45	32	88	61.5	14.06	43	86

GIS, allows water from a given cell to flow into only one adjacent cell, along the direction of steepest descent encoded from 1 to 128 in different directions (Jenson and Domingue, 1988). This difference in elevation is then divided by the center to center distance between directions of water flow shows in Fig. 2. In addition to this, U.S. Salinity Laboratory diagram, Wilcox diagram, Piper diagram, were employed to classify and determine the suitability of groundwater for irrigation by correlating sodium absorption ratio/electrical conductivity and percent sodium/electrical conductivity, respectively (Table 1). WQI was calculated using the World Health Organization standards and Indian Standards. The WQI calculation was carried out using a weighted arithmetic index (Brown et al., 1970).

4. Results and discussion

Results of chemical analysis of physico-chemical parameters of the study area for pre monsoon are shown in Table 2. Statistical summary of physico-chemical parameters and concentrations of ions are compared with World Health Organization (WHO) and Bureau of Indian Standards (BIS) is shown in Table 3.

5. pH

In the study area pH ranged from 7.2 to 8.8 with a mean of 7.9 during pre monsoon. The permissible limit of pH value for drinking water is specified as 6.5 to 8.5 (WHO, 2006; ISI, 1983).

6. Electrical conductivity (EC)

The most desirable limit of EC in drinking water is prescribed as 1500 $\mu\text{mhos}/\text{cm}$ (WHO, 2004; WHO, 1983). Electrical conductivity of the groundwater in the study area is ranging from 410 to 670 $\mu\text{S}/\text{cm}$ with a mean of 529 during pre monsoon (Table 2).

7. Total dissolved solids (TDS)

The concentration of dissolved solids in natural water is usually less than 500 mg/L , while water with more than 500 mg/L is undesirable for drinking and many industrial uses. Water with more than 1000 mg/L of dissolved solids usually gives disagreeable taste or makes the water unsuitable. The high value of TDS influences the taste, hardness, and corrosive property of the water (Hari Haran, 2002; Subhadra Devi et al., 2003). TDS varied from 265 to 435 mg/L with a mean of 338 mg/L during pre monsoon (Table 2). The groundwater in the present study area is of fresh water type according to US Geological Survey (2000) (Table 5).

8. Total hardness (TH)

The acceptable limit of total hardness (TH) (as CaCO_3) is 300 mg/L , which can be extended up to 600 mg/L in case of non-availability of any alternate water source (WHO, 1990). The total hardness as CaCO_3 equivalents in the study area ranges from 360 to 730 mg/L with mean value of 518.8 mg/L (Table 2 and 3). Sawyer et al. (2003) classified groundwater is given in Table 5, according to which 100% of samples belongs to very hard type.

9. Sodium (Na^+) and potassium (K^+)

Sodium concentration in groundwater ranges from 70 to 200 mg/L with a mean of 105 mg/L during pre monsoon season (Table 2). According to WHO (1990) guidelines, the maximum admissible limit is 200 mg/L . All groundwater samples are within the permissible limits. The relative high concentration of sodium may adversely affect soil structure and permeability, resulting in alkaline soils. Higher concentration of sodium in drinking water causes severe health problems like hypertension, congenital diseases, kidney disorders and nervous disorders in human body (Vinod et al., 2009; Narsimha et al., 2017a). The source of sodium (Na^+) is derived geologically from leaching of deposit and decomposition of various minerals. Water in contact with igneous rocks dissolves sodium from its natural source. Sodium may also enter natural waters through industrial, municipal waste discharges and runoff from diffuse sources.

Although potassium is nearly as abundant as sodium in igneous rocks and metamorphic rocks, its concentration in groundwater is one-tenth or even one-hundredth that of sodium. Two factors are responsible for the scarcity of potassium in groundwater, one being the resistance of potassium minerals to decomposition by weathering (Golditch, 1938) and the other being the fixation of potassium in clay minerals are formed due to weathering. Potassium concentration of the groundwater in the study area is ranging from 10 to 36 mg/L . All groundwater samples are within the permissible limits.

10. Bicarbonate (HCO_3^-)

Bicarbonate (HCO_3^-) concentration of the groundwater samples in the study area is ranging from 65 to 130 mg/L during the pre monsoon (Table 2). All groundwater samples lie within the desirable limit of 150 mg/L (WHO, 1990).

11. Calcium (Ca^{2+}) and magnesium (Mg^{2+})

Calcium in the groundwater of the study area is varying from 35 to 80 mg/L with an average value of 53.7 mg/L (Table 2). The permissible limit of calcium in drinking water is 200 mg/L (WHO, 1984). All

Table 4
Dental and skeletal fluorosis affected areas.

	Fluoride concentration	Affected areas	% of the total area
Dental Fluorosis	< 1.5 mg/L	Bramhadevaramarri, Mudigubba, Kondagattupalli, Mangalamadaka, Pulagampalle, Charupalle, Battulapalli and Kadiri	16
	1.5–3.0 mg/L	Dorigallu, Gandlavandlapalli, Gunjepalli, Tappetavaripalli, Ellareddipalli, Marthadu, Sankeppalli, Chagapuram, Sanevaripalli, Chinnakotla, Nallamada, Kurumala, Kadirikuntlapalli, Mutyalacheruvu, Kalasamudram, Eguvapalle, Erradoddi, Patnam, KadiriBrahmanapalli, Pandulakunta, Vankarakunta and Mallepalle	44
	> 3.0 mg/L	Uppalapadu, Nakkalaguttapalle, Malakavemula, Nagireddipalli, Vellamaddi and Gopepalli	12
Skeletal Fluorosis	> 4.0 mg/L	RallaAnantapuram, Timmanayanipalem, Bapanakunta, Budanampalli, Mukthapuram, Kondavandlapalli, Bandlapalli, SankeppalliBrahmanapalli, Jonnalakothapalli, Devaragudipalle, Reddipalle, Donnikota, Alampur, Kavulepalli and Kutagulla	28

groundwater samples are within the permissible limit. Magnesium values varied between 32 and 92 mg/L (Table 2). The required permissible limit of magnesium in groundwater for drinking purpose is 150 mg/L (WHO, 1990).

12. Sulphate (SO_4^{2-}) and chloride (Cl^-)

The Sulphate concentration in the groundwater samples vary from 70 to 240 mg/L with average value of 132 mg/L (Table 3), which is below the permissible limit of 400 mg/L (WHO, 1990). The chloride concentration of the groundwater samples of the study area ranges from 90 to 300 mg/L with an average of 200 mg/L. The desirable limit of chloride in potable water is 200 mg/L, and the permissible limit is 1000 mg/L (WHO, 1990).

13. Nitrate (NO_3^-) and fluoride (F^-)

The concentration of nitrates in groundwater is of primary concern due to potential human health impact from groundwater usage. The upper limit of nitrate concentration in drinking water is specified as 45 mg/L (WHO, 1984). Agriculture and livestock production accounts for nearly 80% of all the nitrogen added to the environment. Nitrate concentration of the groundwater samples in the study area is in the range of 32–88 mg/L during the pre monsoon. Some samples of the north and northeastern part of the study area have high concentration of nitrate. In a small packet of area around the study area, less than 45 mg/L nitrate concentration occurs. It is observed from the study area that 86% of the groundwater samples has exceed limit of nitrate concentration as per WHO standards. High nitrate concentration in groundwater of this area is due to leaching from indiscriminate dumping of animal waste (B.M Reddy et al., 2013).

Fluoride concentrations in the study area varied between 0.9 and

5.7 mg/L 84% of the groundwater samples exceed the permissible limit of fluoride. Dental and skeletal fluorosis affected areas are shown in Table 4. Groundwater is the only source of drinking in these villages. The rate and extent of fluorosis in southeastern part of Anantapur district increases with increase of fluoride level in drinking of water/human. High concentration of fluoride in groundwater is common in the fractured hard rock zone with pegmatite veins (Ramesam and Rajagopalan, 1985). High fluoride level in drinking water causes dental decay and physiological deformations.

Exposure to fluoride beyond the recommended level for longer duration causes fluorosis. Several physical and chemical defluoridation methods have been designed to treat high fluoride concentration in ground water. There is need to develop such methods for defluoridation which would be cost effective and biodegradable. Adsorption studies for defluoridation on *Mentha longifolia* (Mint) showed that the adsorbent were highly influenced by temperature, pH of the solution, and initial fluoride concentration. Experiments at room temperature were performed to evaluate the adsorption capacity of *Mentha longifolia* (Mint) from aqueous solutions. Results show that *Mentha longifolia* (Mint) is as effective for removing fluoride in groundwater samples (Sunitha and MuralidharaReddy, 2018).

High sodium, potassium, bicarbonate and an alkaline pH increase the ion exchange of fluoride by hydroxyl ions (OH^-) The Na^+ acquired by groundwater from the chemical weathering of rock forming minerals is used as an index of the weathering of minerals alternate wet and dry conditions of the semiarid climate. This is responsible for leaching fluoride from the minerals present in the soils and rocks (SubbaRao, 1993). Higher electro negativity of fluoride always facilitates attraction of Ca^{2+} ions in teeth and bones and excess intake of fluoride causes skeletal fluorosis. Fluoride is negatively correlated with calcium which further indicates that there is no substantial contribution of fluoride in increasing its concentration by other anthropogenic sources. Hence

Table 5
Classification of groundwater based on irrigational water quality parameters.

Parameter	Classification	Range	Number of samples	Percentage of samples	References
TDS	Fresh water	< 1000	50	100	US Geological Survey (2000)
	Slightly saline	1000–3000	Nil	Nil	
	Moderately saline	3000–10,000	Nil	Nil	
	High saline	10,000–35,000	Nil	Nil	
TH	Safe	< 75	Nil	Nil	Sawyer et al. (2003)
	Moderate	75–150	Nil	Nil	
	Hard	150–300	Nil	Nil	
	Very Hard	> 300	50	100	
Percent Sodium (%Na)	Safe	< 60	1	2	Wilcox (1955)
	Unsafe	> 60	49	98	
SAR	Excellent	< 10	50	100	Todd (1980)
	Good	10–18	Nil	Nil	
	Fair	18–26	Nil	Nil	
	Poor	< 26	Nil	Nil	
RSC	Good	< 1.25	50	100	Eaton (1950)
	Doubtful	1.25–2.50	Nil	Nil	
	Unsuitable	> 2.50	Nil	Nil	

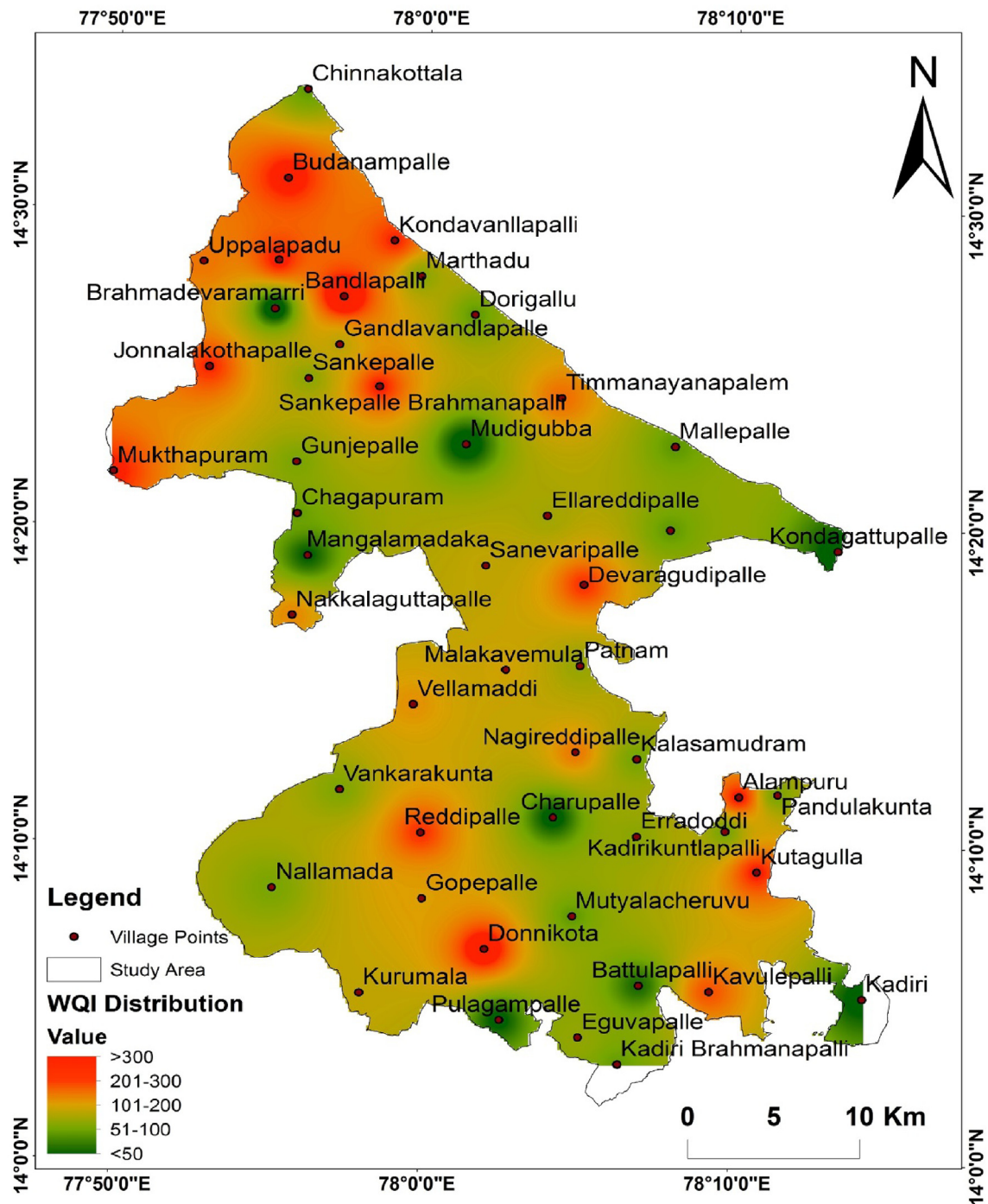


Fig. 3. Spatial Distribution Map of drinking water quality index.

fluoride concentration in groundwater is mainly due to availability and solubility of fluorine-bearing minerals. However in the study area fluoride is not showing significant correlation with pH; whereas positive correlation of fluoride with Na^+ , K^+ , HCO_3^- indicating that the alkaline nature of groundwater with high HCO_3^- and Na^+ can enhance the solubility of fluorite in the groundwater of studied region (Fig. 4 a-e). Correlation coefficient of chemical parameters is shown in Table 9. Further it can also be concluded that F^- contamination of groundwater is a function of many factors such as temperature, pH, concentration of calcium and bicarbonate ions in water, weathering and leaching of fluoride bearing minerals from the basement granitic rocks of this region under alkaline environment.

14. Estimation of water quality index

The evaluation of general suitability of groundwater for drinking purposes for each sample were achieved by estimating their respective water quality index (WQI). WQI was calculated using the World Health Organization standards (WHO, 2011) and Indian Standards (ISI, 2012). The WQI calculation was carried out using a weighted arithmetic index as shown below (Brown et al., 1970).

Calculation for water quality rating:

$$q_n = 100[(V_n - V_i)/(S_n - V_i)]$$

where, q_n = Water quality rating for the nth parameter; V_n = Observed value of the nth parameter, S_n = Standard permissible value of nth

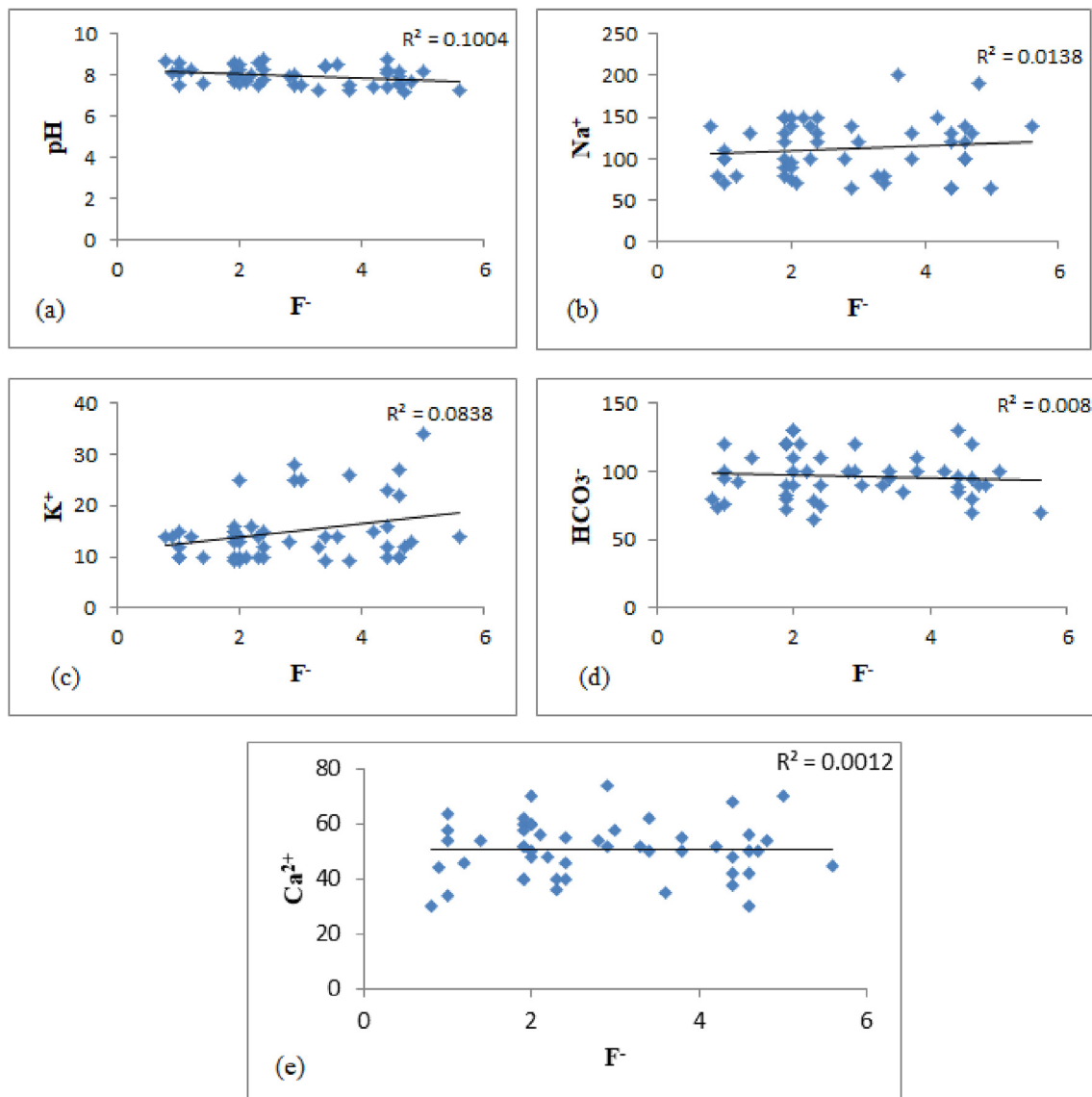


Fig. 4. Fluoride correlation with other physico-chemical parameters of the Groundwater samples.

Table 6

Relative weight for each parameter.

Chemical Parameters	Sn	1/Sn	K	Wn = K/Sn	Ideal Value (Vi)	Observed Value (Vn)	Quality Rating (Qn)	WnQn	WQI
pH	8.5	0.117647	1.187128	0.139662	7	7.5	33.33	4.655	611.4827
EC	1500	0.000667		0.000791	0	470	31.33	0.024	
TDS	1500	0.000667		0.000791	0	300	20.00	0.015	
TH as CaCO ₃	500	0.002		0.002374	0	580	116.00	0.275	
Ca ²⁺	200	0.005		0.005936	0	55	27.50	0.163	
Mg ²⁺	150	0.006667		0.007914	0	32	21.33	0.168	
Na ⁺	200	0.005		0.005936	0	150	75.00	0.445	
K ⁺	200	0.005		0.005936	0	28	14.00	0.083	
HCO ₃ ⁻	150	0.006667		0.007914	0	95	63.33	0.501	
Cl ⁻	600	0.001667		0.001979	0	130	21.67	0.042	
SO ₄ ²⁻	400	0.0025		0.002968	0	150	37.50	0.111	
F ⁻	1.5	0.666667		0.791419	0	4.8	760.00	601.47	
NO ₃ ⁻	45	0.022222		0.026381	0	60	133.33	3.517	
		0.842369		1				611.48	

parameter; V_i = Ideal value of nth parameter.

Calculation of Unit weight (W_n):

The Unit weight of the corresponding parameter was an inverse proportional value to the recommended standard value of S_n

$$W_n = K/S_n$$

where, W_n = Unit weight for the nth parameter; S_n = Standard value of the nth parameter; K = Constant for proportionality.

Constant for proportionality constant by using the following

Table 7

Water quality classification based on WQI value.

Class	WQI Value	Water Quality Status	Percentage
A	< 50	Excellent	10
B	51–100	Good	6
C	101–200	Poor Water	16
D	201–300	Very Poor Water	22
E	> 300	Water Unsuitable for Drinking	46

Table 8

WQI at individual sampling stations.

Sample No	WQI	Water Quality Status
1	611.4827	Water Unsuitable for Drinking
2	482.449	Water Unsuitable for Drinking
3	326.444	Water Unsuitable for Drinking
4	358.1776	Water Unsuitable for Drinking
5	629.1629	Water Unsuitable for Drinking
6	173.1741	Poor Water
7	602.2312	Water Unsuitable for Drinking
8	587.6677	Water Unsuitable for Drinking
9	462.8071	Water Unsuitable for Drinking
10	4.895877	Excellent
11	274.1272	Very Poor Water
12	752.8037	Water Unsuitable for Drinking
13	232.5936	Very Poor Water
14	597.6477	Water Unsuitable for Drinking
15	171.5562	Poor Water
16	13.58391	Excellent
17	44.67325	Excellent
18	239.5298	Very Poor Water
19	178.3008	Poor Water
20	584.0379	Water Unsuitable for Drinking
21	64.16712	Good
22	577.3366	Water Unsuitable for Drinking
23	468.7854	Water Unsuitable for Drinking
24	169.9569	Poor Water
25	330.1727	Water Unsuitable for Drinking
26	173.823	Poor Water
27	207.364	Very Poor Water
28	431.2851	Water Unsuitable for Drinking
29	53.83723	Good
30	577.4571	Water Unsuitable for Drinking
31	203.6058	Very Poor Water
32	331.6874	Water Unsuitable for Drinking
33	682.0551	Water Unsuitable for Drinking
34	414.3402	Water Unsuitable for Drinking
35	387.6101	Water Unsuitable for Drinking
36	40.70402	Excellent
37	216.4783	Very Poor Water
38	202.2105	Very Poor Water
39	172.2913	Poor Water
40	200.4333	Poor Water
41	249.0872	Very Poor Water
42	235.3109	Very Poor Water
43	607.6411	Water Unsuitable for Drinking
44	530.0867	Water Unsuitable for Drinking
45	88.57036	Good
46	597.7201	Water Unsuitable for Drinking
47	45.84956	Excellent
48	254.1926	Very Poor Water
49	177.1507	Poor Water
50	209.1225	Very Poor Water

equation was calculated:

$$K = 1 / \sum (1/S_n)$$

The total water quality index was calculated linearly by adding the quality rating to the unit weight.

$$WQI = \sum q_n W_n / \sum W_n$$

Water quality index was used to assess the status of groundwater

quality for drinking purpose in this area water quality classification based on WQI value is shown in Tables 6–8. Water quality index spatial distribution map shown in Fig. 3. WQI values of groundwater samples ranges from 4.895 to 752.80. Out of fifty samples, only 10% of samples and 6% of samples are of excellent and good category respectively. 16% of samples are of poor category. 22% of samples belong to very poor class and 46% of samples are of unsuitable class. Based on this overall quality of groundwater is not suitable for drinking purpose.

15. Classification of groundwater for irrigation purposes

Irrigation water quality refers to its suitability for agricultural use. According to the U.S. Salinity Laboratory of the Department of Agriculture (1954), electrical conductivity and Sodium adsorption ratios are considered in determining the suitability of water quality for irrigation. The suitability of water for irrigation can be estimated by means of many determinants like Percent Sodium (%Na), Residual Sodium Carbonate (RSC) and Sodium Adsorption Ratio (SAR).

16. Percent sodium (%Na)

The sodium in irrigation waters is usually denoted as percent sodium and can be determined by the following formula given below (Wilcox, 1955). The classification of groundwater samples of the study area are with respect to percent sodium and EC is shown in Fig. 5 for pre monsoon respectively. It is observed that 100% samples during pre monsoon season fall in the excellent to good category.

17. Residual sodium carbonate (RSC)

The sodium hazard to the crops can also increase significantly with the rise in bicarbonate concentration. Water containing high concentrations of carbonate and bicarbonate ions tends to precipitate calcium and magnesium as their carbonates. As a consequence, the relative proportion of sodium increases and gets fixed in the soil thereby decreasing the soil permeability. Residual Sodium carbonate is calculated by the following equation (Eaton, 1950):

$$RSC = [CO_3^{2-} + HCO_3^-] - [Ca^{2+} + Mg^{2+}]$$

where all the concentrations are expressed in meq/L. The groundwater of the study area is classified on the basis of the RSC is presented in Table 2. It is clear from that the groundwater of the area does not have a tendency to cause sodium hazards. The RSC values for all the sampling periods indicate that the groundwater has good quality for irrigation.

18. Sodium adsorption ratio (SAR)

Sodium concentration is an important factor as sodium reacts with soil and reduces its permeability. Therefore, sodium adsorption ratio is considered as a measure of sodium hazard to irrigation. A soil with high exchangeable sodium is very much undesirable for agricultural purposes, become deflocculated and tends to have a relatively impermeable layer. This process of deflocculating is accelerated by water with a high sodium adsorption ratio and reserved by water containing a high proportion of calcium and magnesium (Hem, 1970). The degree to which irrigation water enters into cation exchange reactions in soil can be indicated by SAR. Todd (1980).

$$SAR = \frac{[Na^+]}{\sqrt{\frac{[Ca^{2+} + Mg^{2+}]}{2}}}$$

where ionic concentrations are expressed in milli equivalents. The greater the value of SAR; the greater is the hazard to the crops due to excess sodium. The groundwater of the Mudigubba, Nallamada and Kadiri Mandals is classified on the basis of SAR values as recommended by Todd (1980) and presented in Table 5. It is clear from the table that

Table 9
Correlation coefficient of chemical parameters.

	pH	EC	TDS	TH	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	HCO ₃ ⁻	SO ₄ ²⁻	Cl ⁻	F ⁻	NO ₃ ⁻
pH	1												
EC	0.283	1											
TDS	0.284	0.997	1										
TH	-0.016	0.309	0.303	1									
Na ⁺	0.071	-0.415	-0.418	-0.302	1								
K ⁺	-0.156	-0.075	-0.082	0.178	0.018	1							
Ca ²⁺	-0.072	0.435	0.436	0.571	-0.268	0.216	1						
Mg ²⁺	0.211	0.195	0.201	0.048	-0.104	0.104	0.232	1					
HCO ₃ ⁻	0.019	0.448	0.444	0.371	-0.051	-0.081	0.572	0.048	1				
SO ₄ ²⁻	-0.019	-0.015	-0.016	-0.024	0.351	0.067	0.294	0.158	0.343	1			
Cl ⁻	-0.078	0.311	0.311	-0.078	-0.179	-0.232	0.246	0.323	0.436	0.471	1		
F ⁻	0.284	-0.068	-0.076	0.096	0.108	0.326	0.0007	-0.158	-0.084	0.022	-0.204	1	
NO ₃ ⁻	0.399	0.359	0.355	0.102	-0.148	-0.027	0.102	-0.155	-0.039	-0.397	-0.518	0.056	1

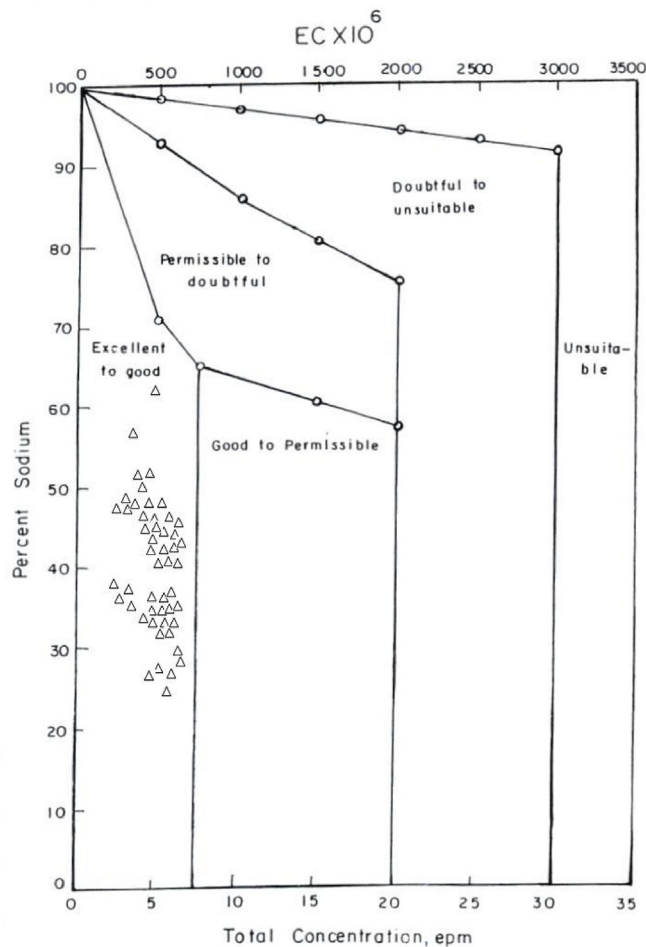


Fig. 5. Classification of groundwater with respect to percent sodium and EC (Wilcox, 1955).

SAR of the 100% of groundwater samples in study area belongs to excellent class.

The graphical method proposed by Raghunath (1987), is used to examine the suitability of groundwater for irrigation. It is essentially an adaptation of the semi-logarithmic diagrammatic representation developed by the staff of U.S. Salinity laboratory (1954), to suit Indian conditions. The diagram uses two parameters namely, (i) Electrical conductance (EC) of water expressed in $\mu\text{S}/\text{cm}$, which is proportional to TDS content and hence the degree of salinity and (ii) Sodium adsorption ratio (SAR). Irrigation water classification based on SAR indicates that 100% of the samples are suitable for irrigation (Fig. 6).

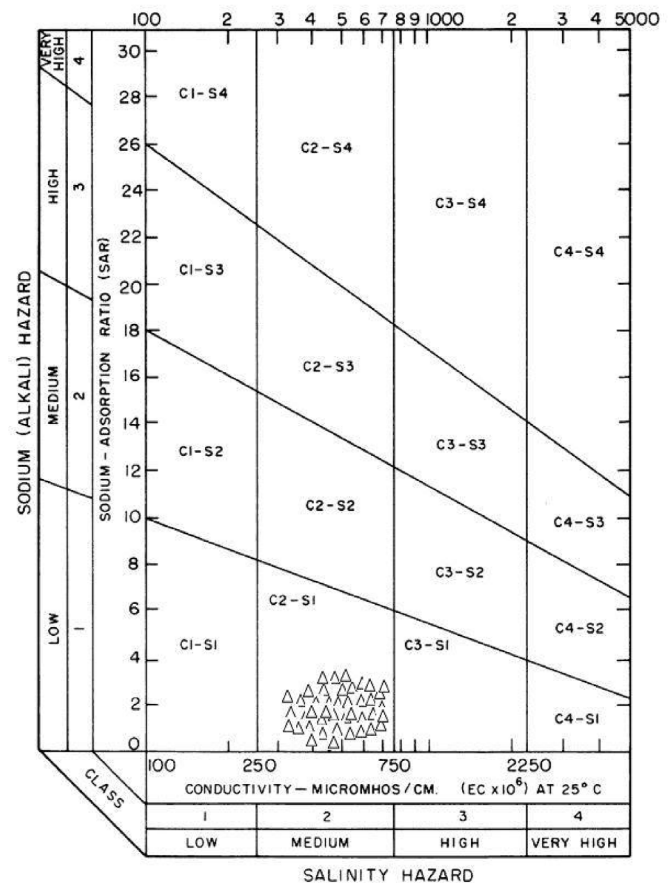


Fig. 6. USSL Classification of groundwater with respect to SAR and EC.

19. Piper Tri-linear diagram

The Piper trilinear diagram (Piper, 1953) for the groundwater samples is presented in Fig. 7, which clearly explains the variations of cation and anion concentration in the study area. The diamond-shaped field of piper diagram is further divided into four classes, namely $\text{Ca}^{2+}\text{-Mg}^{2+}\text{-Cl-SO}_4^{2-}$, $\text{Na}^{+}\text{-K}^{+}\text{-Cl-SO}_4^{2-}$, $\text{Na}^{+}\text{-K}^{+}\text{-HCO}_3^{-}$ and $\text{Ca}^{2+}\text{-Mg}^{2+}\text{-HCO}_3^{-}$. In the study area majority of samples belong to the $\text{Na}^{+}\text{-K}^{+}\text{-Cl-SO}_4^{2-}$ type water.

20. Conclusion

The main objective of the study was to assess the groundwater quality for drinking & irrigation purpose. The groundwater in the study area is alkaline in nature. Groundwater is exceeding the permissible

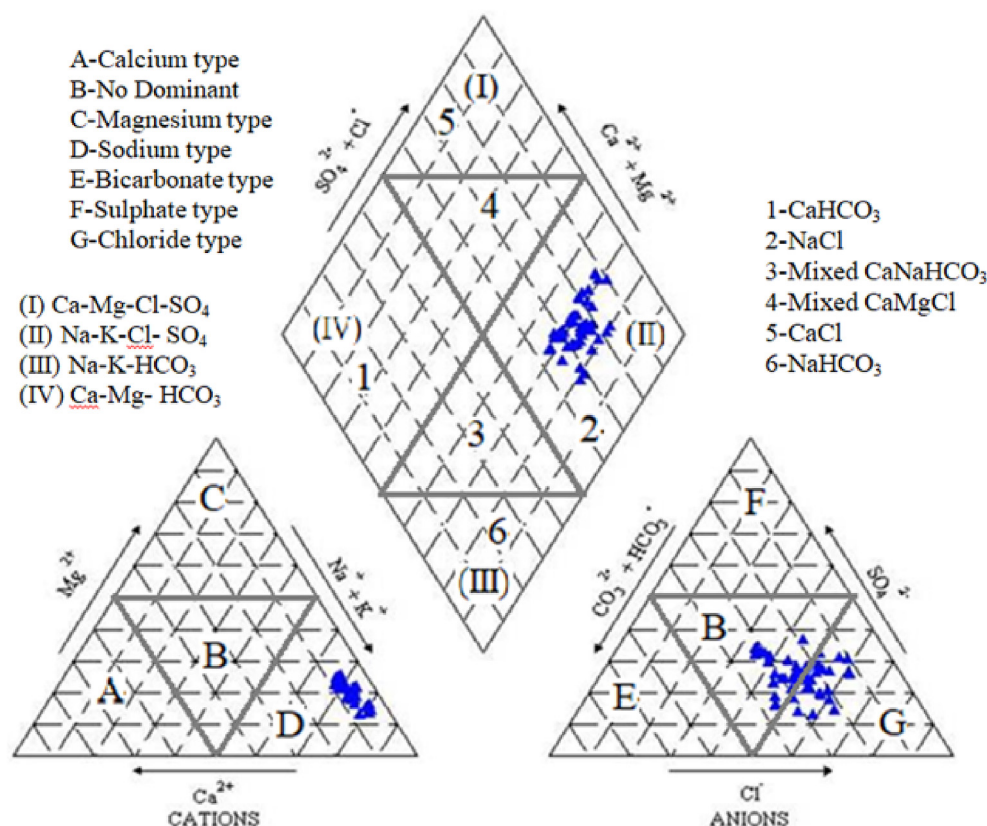


Fig. 7. Piper plot describing hydrogeochemical facies of the study area.

limit of pH, Total hardness, nitrate and fluoride. Cases of dental and skeletal fluorosis are observed in this area due to continued intake of water enriched with fluoride. Possible source of fluoride in groundwater are weathering and leaching of fluoride bearing minerals from the basement granitic rocks of this region under alkaline environment. More than 80% of the samples exceed the permissible limit (1.5 mg/L) of fluoride and proper defluoridation techniques are necessary to purify this water. On the basis of WQI values of groundwater samples ranges from 4.895 to 752.80. Out of fifty samples, only 10% of samples and 6% of samples are of excellent and good category respectively. 16% of samples are of poor category. 22% of samples belong to very poor class and 46% of samples are of unsuitable class. According to Sawyer et al. (2003) classification, all the groundwater of the present study area is rated as very hard during pre monsoon season and requires processing before use. The suitability of water for irrigation can be estimated by means of many determinants like Percent Sodium (%Na), Residual Sodium Carbonate (RSC) and Sodium Adsorption Ratio (SAR). Sodium adsorption ratio (SAR) values are to be less than 10 and belong to excellent class during pre monsoon season and are suitable for irrigation. Residual sodium carbonate (RSC) values on the whole are less than 1.25 meq/L; Most of the water samples of the study are good for irrigation. According to the as per USSL diagram all water samples are falling under medium specific conductance and low sodium (C_2S_1) and are suitable for irrigation. The Wilcox classification has shown 100% of groundwater under excellent to good zone. Thus, the overall groundwater quality in the basin is not desirable for drinking and is suitable only for irrigation purpose.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gsd.2019.100262>.

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